CPT208 Human-Centric Computing

X-thread

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Introduction

Problems

At XJTLU, group discussions are critical to academic success—but students face persistent barriers: conflicting schedules make synchronous collaboration hard to coordinate; on-campus study spaces are often unavailable; unfamiliar team members make it difficult to break the ice and thus hinder the progress of the discussion; real-time academic support is missing, slowing progress on complex questions; and key insights are easily lost without structured documentation, leaving teams with disorganized notes and weak follow-up. These issues lead to fragmented, unproductive discussions and poorer learning outcomes.[1]

Opportunities

Advancements in GenAI and collaborative digital tools offer clear solutions: GenAI provides instant, on-demand academic guidance during discussions, while shared workspaces enable real-time mind mapping, whiteboarding, and idea documentation. Together, these technologies eliminate location and scheduling limits, while directly addressing gaps in support and post-meeting retention.[2][3]

Aims

- Break Geographical & Temporal Barriers—Empower XJTLU students with a flexible, hybrid collaboration platform that eliminates scheduling and location barriers, enabling them to join group discussions anytime, anywhere.
- Deliver Real-Time AI-Powered Support—Provide instant, context-aware academic support through integrated GenAI, helping students resolve questions and clarify concepts in real time to keep discussions productive.
- Preserve & Organize Key Insights—Enhance idea retention and post-meeting accountability with collaborative mind mapping and shared whiteboarding tools, ensuring key insights are captured, organized, and easily revisited.
- Streamline & Boost Participation — Eliminate awkward silences and low engagement in randomly-assigned teams with guided onboarding and structured workflows, reducing social friction and helping groups dive into productive discussions faster.

Requirements Discovery

Interview:

1. How do you usually collaborate on group projects at XJTLU? What are the most frustrating pain points you face?

Usually, the time is first determined in the group chat, and then a meeting is arranged in the offline study room for concentrated discussion. When the time cannot be arranged, communication is conducted via voice messages on WeChat, and the progress is followed up through group chat records and manual notes throughout the process.

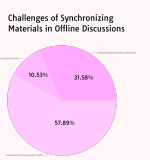
The course schedule is severely conflicting. The offline classrooms are often fully occupied. Many discussions have to be postponed until late at night and the group members are not familiar with each other, resulting in an awkward atmosphere. Sometimes it is difficult to start the topic, leading to low efficiency. The group chat information is chaotic and disorderly, and key viewpoints can easily be deleted. Each meeting requires repeated review, resulting in a lot of ineffective communication.

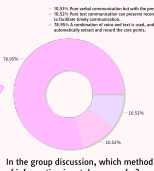
2. When you get stuck on a complex academic question during group work, how do you usually handle it? How effective is your current approach?

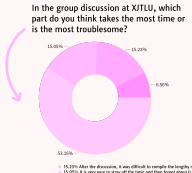
Most of the time, I would first study the materials by myself and fully understand them, then share the knowledge with my group members. If everyone still couldn't grasp it, we would have to temporarily put it aside and wait until after class to ask the teacher or seek help from senior classmates.

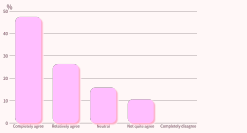
There was no immediate academic support throughout the process, which made it easy to get stuck on the same issue repeatedly and waste time; the manually compiled notes were scattered and disorganized, making it impossible to find the discussion logic at that time when reviewing later, resulting in difficulties in subsequent progress.

Questionnaire:

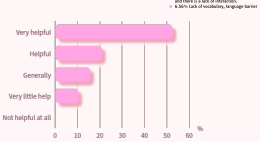






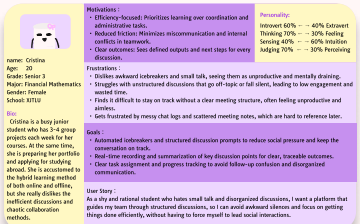


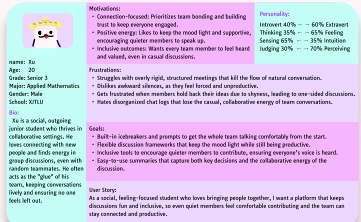
During the first group discussion with unfamiliar classmates, the ice-breaking process was quite awkward and it was difficult to immediately enter a highly productive and active discussion state.



Would it be helpful for your group if we could use "dynamic mind maps" to connect the main points while you are discussing?

Target Users:





Key Requirements:

1、Structured Onboarding & Icebreaking Support

Offer dedicated team matching and guided icebreaker activities to help users quickly adapt to unfamiliar teammates and reduce awkwardness

2、Live AI-Powered Problem Solving

Provide real-time, in-meeting AI assistance to answer study-related questions and resolve confusion on the spot, preventing wasted time due to roadblocks.

3、Collaborative Mind Mapping & Idea Organization

Support live co-editing of mind maps during discussions to capture key insights in real time, with automatic post-meeting summaries for easy review.

4、Quick & Flexible Idea Logging

Include a shared note-taking board to capture spontaneous thoughts, ensuring no ideas are lost during the discussion.

5、Seamless Document Sharing & Analysis

Enable in-meeting file sharing and collaborative annotation for efficient document review and discussion, supporting post-meeting summary work.

6、Full-Featured video Call Environment

Offer a stable, full-featured video call experience to support smooth communication for hybrid team collaboration.

Conclusion

Summary：

The development of the X-Thread APP is mainly aimed at providing students with a meeting platform that enables them to collaborate in groups at any time. By integrating functions such as real-time whiteboard, mind mapping, AI guidance, and ice-breaking interaction, it transforms scattered discussions into structured and synchronized collaborative processes, significantly improving the efficiency of team communication and the participation experience.

Limitations：

1. Currently, the language function only supports Chinese and English. However, the users come from various countries, and the language version has its limitations.
2. However, after the meeting, there is no summary of each person's participation level, and it is impossible to automatically calculate the contribution of each member to the meeting.

Future：

1. Add more language versions to better suit students from different countries, making the app more universal, practical and accessible.
2. Add a post-meeting summary function to analyze each member's participation, helping users review their performance and improve their teamwork skills.

Design and Methodology

Design Alternatives:

1. Comparison of Discussion Framework Schemes

Unstructured Free Discussion: Basic video call only, no agenda/guidance.
✓ Pros: Full freedom for natural conversation
✗ Cons: Prone to silences, off-topic tangents, low introvert engagement;

Semi-Structured Template Discussion: Pre-built templates (icebreakers, agendas) to guide flow.
✓ Pros: Fast ice-breaking, focused discussions, reduced chaos
✗ Cons: Inflexible, rigid for extroverts, limits creativity

AI-Powered Hybrid Mode: Structured guidance + AI Icebreakers/assistance + flexible free discussion + customizable AI pet companions.
✓ Pros: Balances efficiency (introverts) & atmosphere (extroverts), cuts collaboration friction; AI pets enable personalized, low-pressure ice-breaking, atmosphere adjustment and topic initiation, further enhancing engagement for all user types
✗ Cons: Higher development complexity, requires refined UX design for pet interaction and AI prompt customization

Final Choice: AI-Powered Hybrid Mode
Selected via user research, balancing structure, ice-breaking, and creativity for all user personalities, with the added value of customizable AI pets to solve awkward silences, boost group atmosphere, and meet personalized interaction needs.

2.Comparison of Platform Options

Mobile-only Version: App optimized for smartphones only
✓ Pros: High portability, easy access anytime/anywhere
✗ Cons: Small screen limits mind map co-editing & file analysis, poor multitasking experience

Desktop-only Version: Software optimized for PCs only
✓ Pros: Large screen for full-featured collaboration, stable performance
✗ Cons: Low mobility, inconvenient for on-the-go quick discussions

Cross-Platform Hybrid Version: Full sync between mobile & desktop
✓ Pros: Combines mobile portability and desktop productivity, seamless data sync across devices
✗ Cons: Higher development & maintenance cost, but optimal user experience

Final Choice: Cross-Platform Hybrid Version
Selected based on user needs for both mobile convenience and desktop collaboration, ensuring full functionality across all usage scenarios.

3.Comparison of Solutions after Collaboration and Meetings

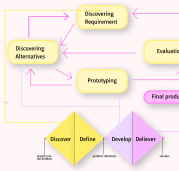
Basic video call only: Only basic audio & video functions, no auxiliary tools.
✓ Pros: Simple operation, high user familiarity.
✗ Cons: No idea organization, recording or AI support; low efficiency, scattered post-meeting information.

Video call + basic sharing: Adds note board, file sharing and mind mapping on top of video calls.
✓ Pros: Supports idea recording, mind mapping and document collaboration.
✗ Cons: No AI Q&A, cannot solve problems instantly during discussions.

Integrated full collaboration: Combines video conferencing, real-time AI Q&A, co-editing mind map, shared board, file sharing and AI meeting summary.
✓ Pros: Enables instant problem-solving, real-time idea sorting and one-stop post-meeting review.
✗ Cons: Higher development cost but best user experience.

Final choice: Integrated full collaboration
Covers the full workflow from in-meeting collaboration to post-meeting organization, greatly improving team efficiency.

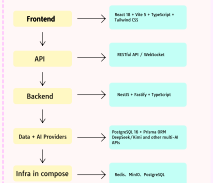
Interaction Design Lifecycle Model:



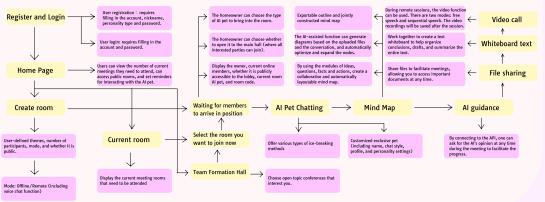
Discovering Requirements: Identifying the core pain points of group collaboration among students at XJTLU University (lack of interaction, low efficiency, scattered information), through research, the Cristina & Xu dual user profile was produced, and 6 core requirements were determined.
Designing Alternatives: Producing 3 sets of solution options for the identified pain points (discussion framework, collaboration function, platform selection), and finally selecting the AI-powered hybrid model + fully functional integrated solution.
Prototyping: Implementing as flowcharts and high-fidelity prototypes, covering all functions such as video meetings, AI Q&A, AI pets, mind maps, and file sharing.
Evaluating: Through task testing, user interviews, and heuristic evaluation, 3 rounds of iterative optimization were conducted, ultimately producing a product that meets user needs.

Process of three iterations:

Function / Iteration	AI Pet Icebreaker	Collaborative Thinking Tools	Meeting & Collaboration Enhancement
Iteration I	There is no ice-breaking function.	Provide the shared whiteboard function to help record temporary ideas, organize conclusions, drafts and collaborative notes.	Launch basic video conference service; support creation and opening of basic rooms, meeting the basic requirements for remote communication.
Iteration II	Only provide basic fixed templates for pets. The pets will generate simple ice-breaking topics for use at the beginning of meetings to create a warm atmosphere.	Add the basic collaborative editing mind mapping function, which helps to organize issues, ideas, facts, actions, and enables the function of automatically arranging the chaotic mind maps in one click.	Add file transfer sharing and team lobby functions, support dynamic recommendation of users based on themes and skills, and achieve precise matching of like-minded individuals.
Iteration III	Newly added pet customization (personality, exclusive dialogues). Pets generate personalized icebreakers based on meeting themes, member viewpoints and organizer expectations, and adjust atmosphere in real time.	Integrate AI capabilities to support automatic organization of meeting ideas, generation of diagram frameworks, optimization of structure, and one-click export functions, thereby enhancing collaboration efficiency.	Integrate multiple AI APIs to enhance the real-time answering capability and solve the problem of discussion stagnation. Improve the video conference and add the function of recording and saving.



System Architecture and Key Features:



Evaluation Methods：

In order to evaluate the availability and user experience of x-thread, We invited several students from XJTLU to go through a series of steps from registration to the simulation meeting, making extensive use of our products. After the experience, the following three testing methods were adopted:

1. System Usability Scale (SUS)
We once again adopted the standardized 10-question questionnaire format, and had 15 XJTLU students who had experienced our program fill it out. Through a 0-100 point scoring system, it reflected the users' overall perception of the system's usability and learnability.

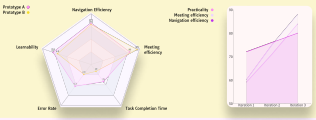
2. Heuristic Evaluation
The evaluator reviewed the interface design and interaction logic based on the ten usability principles of Nielsen, identifying potential usability issues.

3. A/B Testing
We conducted a comparative test on two interaction designs for the ice-breaking function of the AI intelligent pet, aiming to optimize the user experience and the meeting process.

Feedback：

1. SUS score: 82/100, indicating that users highly rated the usability and learnability of the system.
- 85% of the users considered the product instructions clear, the interface neat, and easy to use.
- 90% of the users did not encounter major bugs during use, demonstrating the excellent stability of the product.
- 80% of the users utilized the programs within the product during use, reflecting the practicality of the product.

Result: After feedback, we modified the meeting UI interface of the product, moving the function buttons to the top, leaving more space to facilitate users' message discussion, information search.



Evaluation and Impact

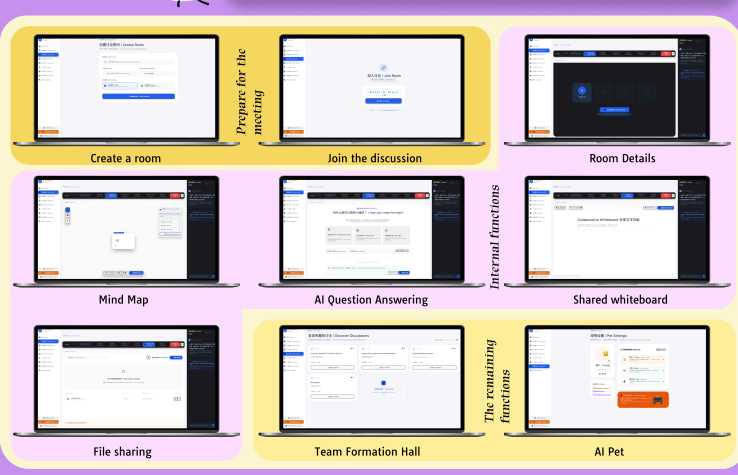
2. Heuristic Evaluation:
- System Status Visibility
Clear status indications are provided for file transfer, voice conferences, and real-time collaboration.
Users can always know the current progress, the online status of members, and the synchronization status.
- Recognition Rather Than Recalling
Core functions (mind maps, whiteboards, files, voice) are directly visible on the interface.
Using common icons and a clear layout, users do not need to remember the operation paths.
- Consistency and Standardization
A unified visual style, button positions, and interaction logic are applied across all platforms.
It conforms to the usage habits of common collaboration software, reducing learning costs and minimizing operational errors.[4]

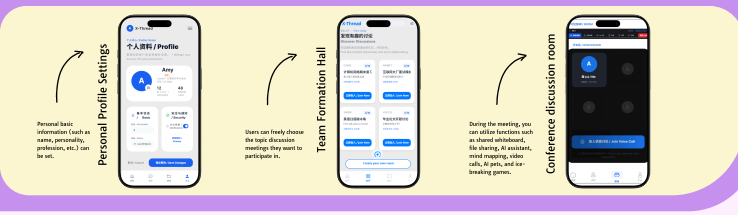
3. A/B Testing
1. Prototype A (Manual Trigger Mode)
Users actively ask questions or make requests by @ing the pet's name to trigger ice-breaking topics or mini-games, which can be paused at any time.
2. Prototype B (Automatic Trigger Mode)
The pet automatically pops up interactive content at fixed times or when it is silent, without the need for manual operation.
Through the simulation meeting test, the performance of the two prototypes, A and B, in terms of meeting efficiency, navigation efficiency, ease of learning, error rate, etc. was compared: Prototype B has stronger interactivity, but it is prone to interrupting discussions and distracting users' attention; Prototype A, on the other hand, enables users to control the pace independently, avoiding interference with the focus of the meeting. Therefore, Prototype A was ultimately selected as the AI pet prototype.

Impact:

X-Thread uses real-time whiteboarding, mind mapping and AI guidance to visualize discussions and improve collaboration. It lowers social and technical barriers for inclusive participation, solves remote meeting fragmentation, and turns inefficient communication into structured, recordable output.

Implementation and Outcomes





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